# Syllabus: Dynamical Systems in Neuroscience

## Textbooks

- Izhikevich, E.M. *Dynamical Systems in Neuroscience: The Geometry of Excitability and Bursting* (MIT Press, 2007)
- Strogatz, S.H. *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering* (3rd Edition, 2018)

## Week 0: Why We're Interested in Dynamical Systems and Self-Study Preparatory Module

- **Objective:** This first read gives a general, high-level overview of topics of interest in dynamical systems for neuroscience. The additional optional material is designed to ensure all participants have the foundational computational skills needed for the course. Since most participants may already be familiar with these tools, these resources are provided as a refresher or a starting point for those who need it.
- **Recommended Resources:**
  - **1. Introductory read:** Miller, Dynamical systems, attractors, and neural circuits (2016).[1]
  - **2. Setting up a Python Environment:**
    - See slides from the coding bootcamp we ran at the beginning of the academic year – ask Nicolas if you need them.
  - **3. Refresher on Linear Algebra Concepts & Practice:**
    - **Conceptual Intuition:** For a highly visual and intuitive understanding of core linear algebra concepts, see the **"Essence of Linear Algebra"** series by 3Blue1Brown.[2]
    - **Foundational Topics: Udemy** is free for anyone at the Crick and offers a comprehensive module on Linear Algebra covering vectors, matrices, and transformations.[3,4]
    - **Implementation with NumPy:** The numpy library is the foundation for numerical computing in Python. Review how to create arrays/vectors, perform matrix multiplication (np.dot or np.matmul), find the inverse (np.linalg.inv), and solve systems of linear equations (np.linalg.solve).[5,6,7,8]
  - **4. Introduction to Scientific Plotting and ODE Simulation:**
    - **Plotting with Matplotlib:** Familiarize yourself with the basics of creating plots using matplotlib. The standard workflow involves creating a figure and axes (fig, ax = plt.subplots()) and then using methods like ax.plot() to visualize data.[9,10,11]
    - **Simulating ODEs with SciPy:** The scipy.integrate.solve_ivp function is the modern tool for solving ordinary differential equation initial value problems in Python. Review how to define a function for your system of ODEs, set the time span and initial conditions, and call the solver to get a solution.[12,13,14]

**Part I: Single Neuron Dynamics**
**Week 1: Introduction to Dynamical Systems**
- **Readings:** Strogatz Ch. 1; Izhikevich Ch. 1.
- **Topics:** What is a dynamical system? State space, trajectories, phase portraits. The neuron as a dynamical system.
- **Computational exercise:** Plotting a simple vector field for a 1D system.

**Week 2: One-Dimensional Flows and Neuronal Excitability**
- **Readings:** Strogatz Ch. 2; Izhikevich Ch. 2.
- **Topics:** Fixed points, stability, and graphical analysis. Neuronal excitability and the leaky integrate-and-fire model.
- **Computational exercise:** Writing a function to find fixed points of a 1D system numerically.

- **Week 3: Bifurcations in One Dimension and Neural Thresholds**
- **Readings:** Strogatz Ch. 3; Izhikevich Ch. 3.
- **Topics:** Saddle-node, transcritical, and pitchfork bifurcations. The saddle-node bifurcation as a model for spike generation. Bistability and hysteresis.
- **Computational exercise:** Simulating a saddle-node bifurcation and plotting the bifurcation diagram.

**Week 4: Two-Dimensional Linear Systems**
- **Readings:** Strogatz Ch. 5; Izhikevich Ch. 4.
- **Topics:** Linearization at a fixed point. Classification of fixed points (saddles, nodes, spirals) using eigenvalues.

**Week 5: Two-Dimensional Nonlinear Systems and Phase Plane Analysis**
- **Readings:** Strogatz Ch. 6; Izhikevich Ch. 5.
- **Topics:** Nullclines and geometric analysis. The FitzHugh-Nagumo model. Phase plane analysis of the Hodgkin-Huxley model.
- **Computational exercise:** Plotting nullclines and trajectories for the FitzHugh-Nagumo model.

**Week 6: Limit Cycles, Oscillations, and Bifurcations in 2D**
- **Readings:** Strogatz Ch. 7 & 8; Izhikevich Ch. 6 & 7.
- **Topics:** Poincaré-Bendixson Theorem. Hopf bifurcations and the onset of oscillations. Homoclinic bifurcations. Firing patterns as bifurcations.

**Part II: Neural Populations Dynamics**

**Week 7: Attractor dynamics in populations of neurons**

- **Readings:** Line attractor dynamics in the SC (Seung 1996)[15]. Ring attractor dynamics in flies (Angelaki and Laurens 2019)[16].
- **Topics:** From two-dimensional dynamical systems to systems of neurons. Attractor models in small populations of neurons.

**Week 8: The State-Space Perspective, Dimensionality Reduction and Neural Manifolds**

- **Readings:** Tutorial on PCA[17]. Saxena & Cunningham (2019)[18]. Gallego et al. (2017)[19].
- **Topics:** Representing population activity in an N-dimensional state space. The motivation for dimensionality reduction. Conceptual and mathematical introduction to PCA. The concept of a neural manifold.
- **Computational exercise:** Performing PCA on a sample (neurons x time) data matrix and visualizing the top principal components.

**Week 9: Foundational Observations in Population Dynamics**

- **Readings:** Churchland et al. (2012)[20]. Vyas et al. (2020)[21].
- **Topics:** Rotational dynamics in motor cortex. The "dynamical systems hypothesis" vs. the "representational view."

**Week 10: Modeling Dynamics: Recurrent Neural Networks (RNNs)**

- **Readings:** Sussillo (2014)[22]. Durstewitz et al. (2023)[23].
- **Topics:** RNNs as models of neural circuits. Attractor dynamics. Reconstructing dynamics from data with RNNs.
- **Computational exercise:** Simulating a simple rate-based RNN and observing its evolution towards a fixed-point attractor.

**Week 11: The Dynamics of Cognition: Decision-Making and Working Memory**

- **Readings:** Wong & Wang (2006)[24].
- **Topics:** Attractor models of decision-making. Line attractors for working memory. The role of dynamics in evidence integration.

**Week 12: The Dynamics of Learning**

- **Readings:** Sussillo & Abbott (2009)[25].
- **Topics:** Hebbian plasticity in dynamical systems. FORCE learning in recurrent networks. Dynamic and plastic attractor landscapes.

**Week 13: Chaos and the Brain**

- **Readings:** Skarda & Freeman (1987)[26]. Korn & Faure (2003)[27]. Terada & Toyoizumi (2024)[28].
- **Topics:** How the brain balances order and disorder for optimal computation. Evidence for chaos in the brain. Modern perspective on how chaos enables complex computations like Bayesian inference.

## References


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